



Operators in C

Master the Building Blocks of C Programming



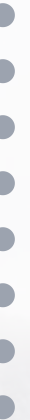
Assignment



Arithmetic



Relational





Introduction



This unit introduces **variables, constants, datatypes** and how to declare them in C programming.

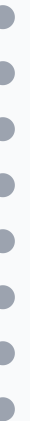


An **expression** is a sequence of operators and operands that:

-  Specifies the computation of a value
-  Designates an object or function
-  Generates side effects



An **operator** performs operations on **operands**



? What are Operators?



Definition

A symbol used to perform operations on variables, operands, or constants



Types

Some require 2 operands (binary), some require 1 (unary)



Categories of Operators

=

Assignment

+ - * / %

Arithmetic

< > =

Relational

&& || !

Logical

++ --

Increment/Decrement

?:

Conditional

,

Comma

sizeof

Size of



= Assignment Operators

Used to **assign values** to variables from right side to left side

```
Variable = constant / variable / expression;
```

Example:

```
#include<stdio.h>
void main() {
    int i;
    i = 5;
    printf("%d", i); // Output: 5
    i = i + 10;
    printf("\n%d", i); // Output: 10
}
```

Shorthand Assignment Operators

`i += 5` → `i = i + 5`

`i -= 5` → `i = i - 5`

`i *= 2` → `i = i * 2`

`i /= 6` → `i = i / 6`

+ Arithmetic Operators

Basic Operations

+

Addition

-

Subtraction

×

Multiplication

÷

Division

%

Modulus (remainder)

Example (a=25, b=4)

a + b = 29

a - b = 21

a * b = 100

a / b = 6 (decimal truncated)

a % b = 1



Important: Division by zero is undefined and causes runtime error!



Relational Operators

Used to **compare two operands** and return true or false

==

Equal to

<

Less than

!=

Not equal to

>= Greater than or equal

>

Greater than

<= Less than or equal

Example Usage

```
if (a <= b) {  
    printf("%d <= %d\n", a, b);  
} else {  
    printf("%d > %d\n", a, b);  
}
```



Code Examples

Arithmetic Operations

```
#include<stdio.h>
main() {
    int a = 21;
    int b = 10;
    int c;

    c = a + b; printf("c = %d\n", c); // 31
    c = a - b; printf("c = %d\n", c); // 11
    c = a * b; printf("c = %d\n", c); // 210
    c = a / b; printf("c = %d\n", c); // 2
    c = a % b; printf("c = %d\n", c); // 1
}
```

Relational Operations

```
#include<stdio.h>
main() {
    int a = 21, b = 10;

    if (a == b) printf("a is equal to b\n");
    else printf("a is not equal to b\n");

    if (a < b) printf("a is less than b\n");
    else printf("a is not less than b\n");

    if (a > b) printf("a is greater than b\n");
}
```

```
else printf("a is not greater than b\n");  
}
```



Summary

=

Assignment

Store values in variables

+ - * / %

Arithmetic

Perform calculations

< > =

Relational

Compare values



Master these operators to build powerful C programs!



